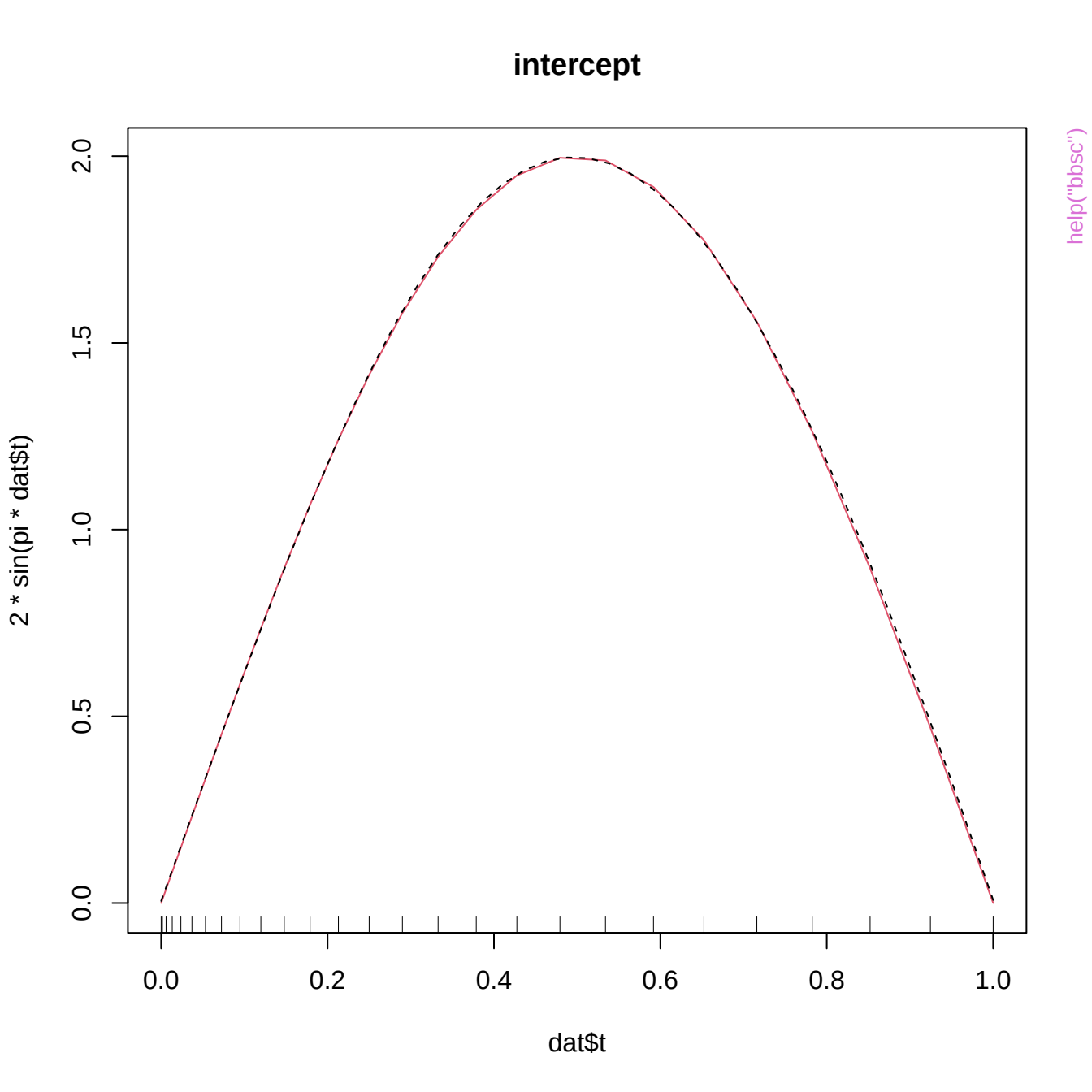
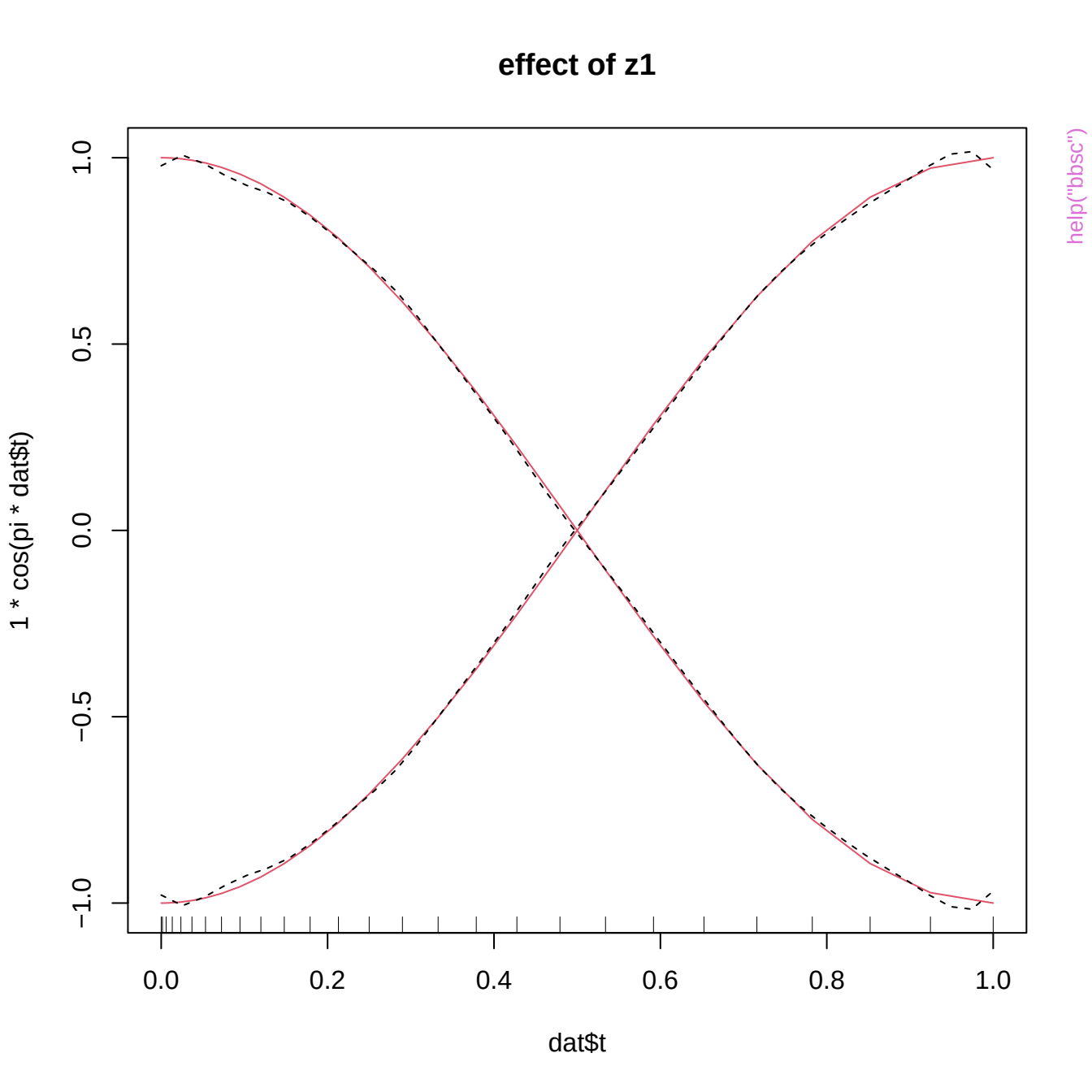
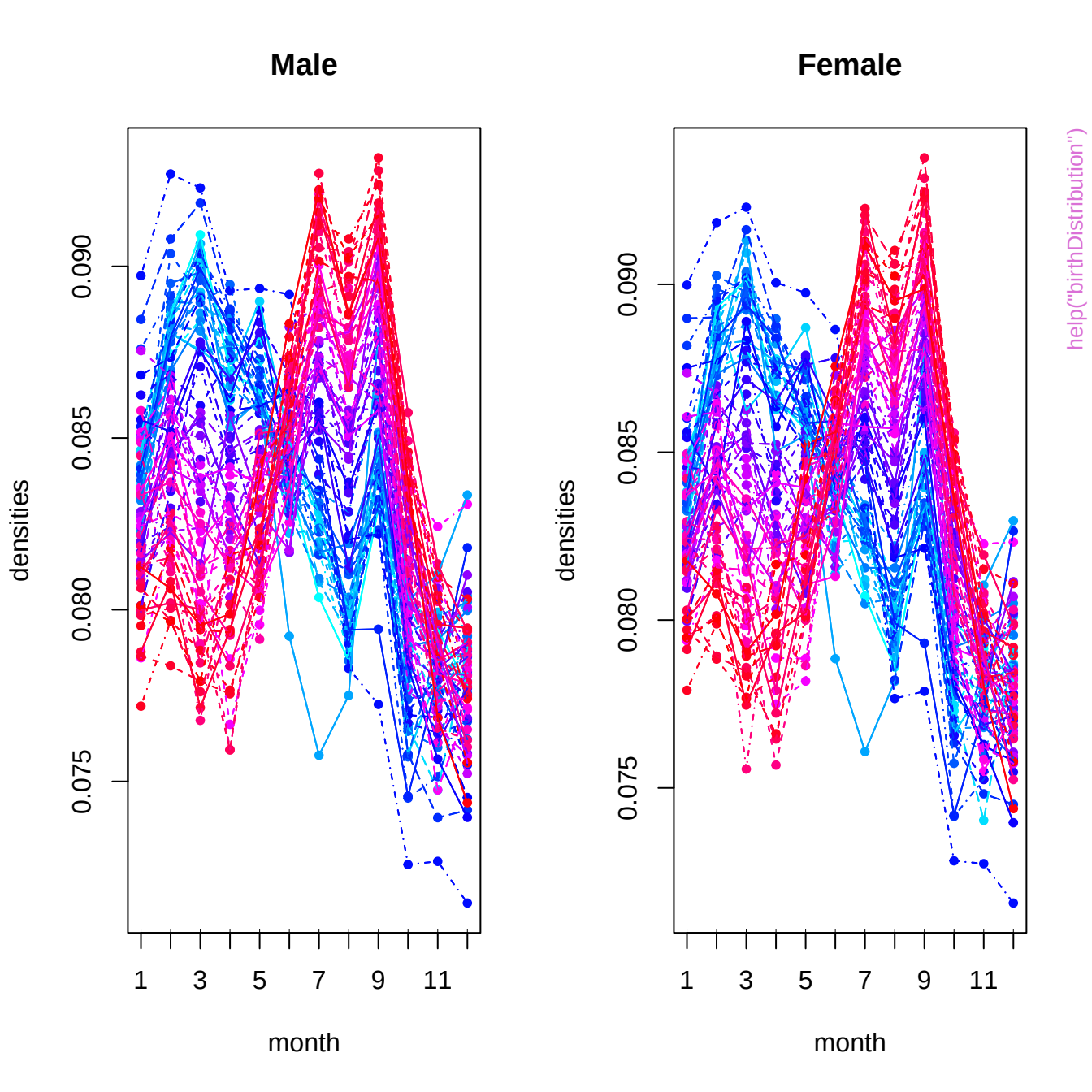
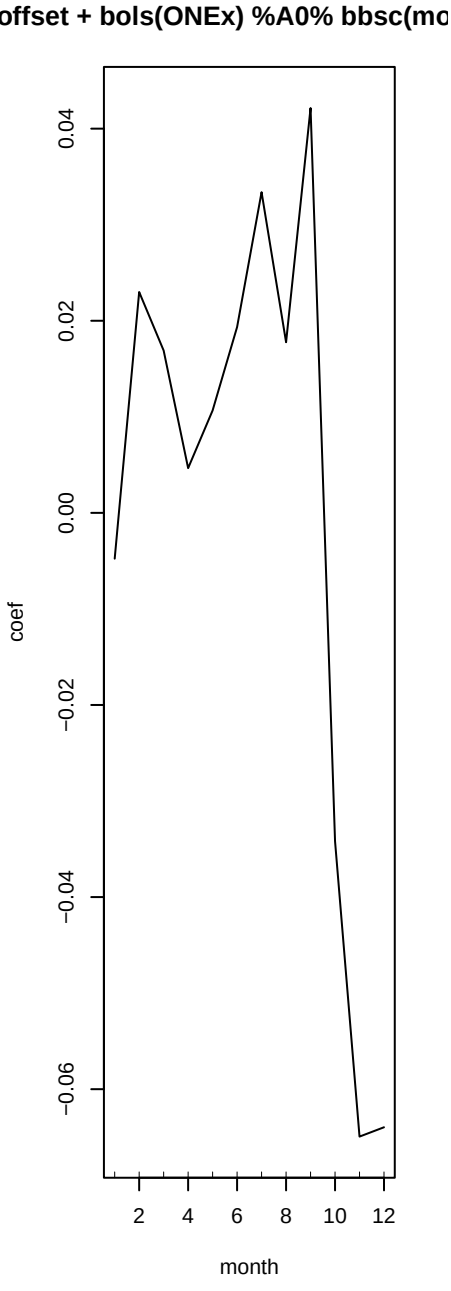
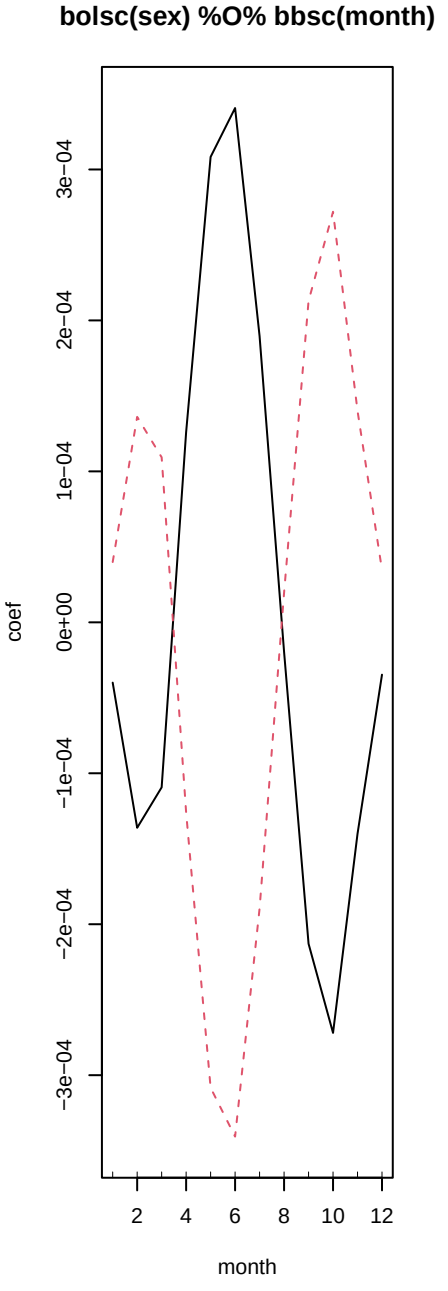
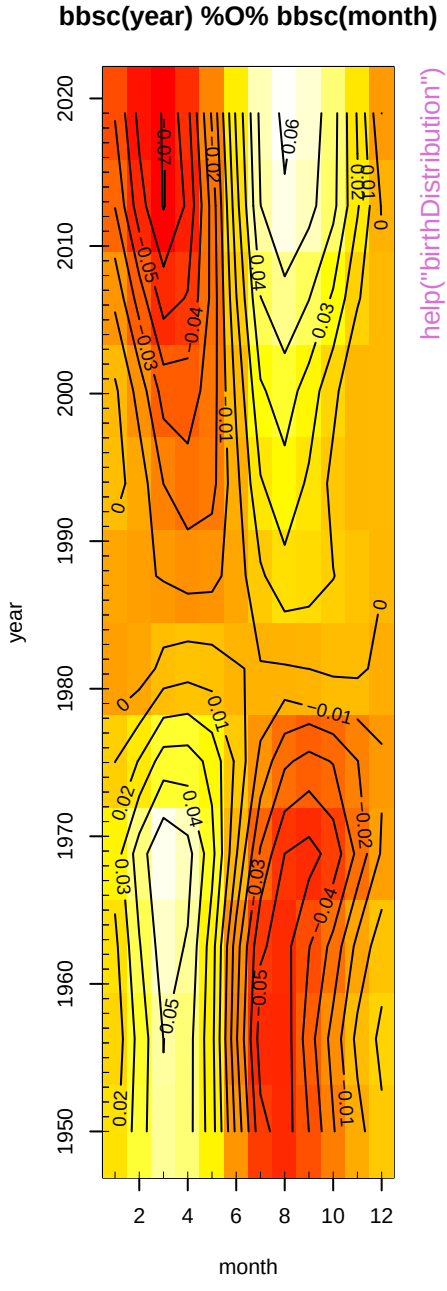

intercept
2 * sin(pi * dat$t)
dat$t
0.0
0.5
1.0
1.5
2.0
0.0
0.2
0.4
0.6
0.8
1.0
help("bbsc")

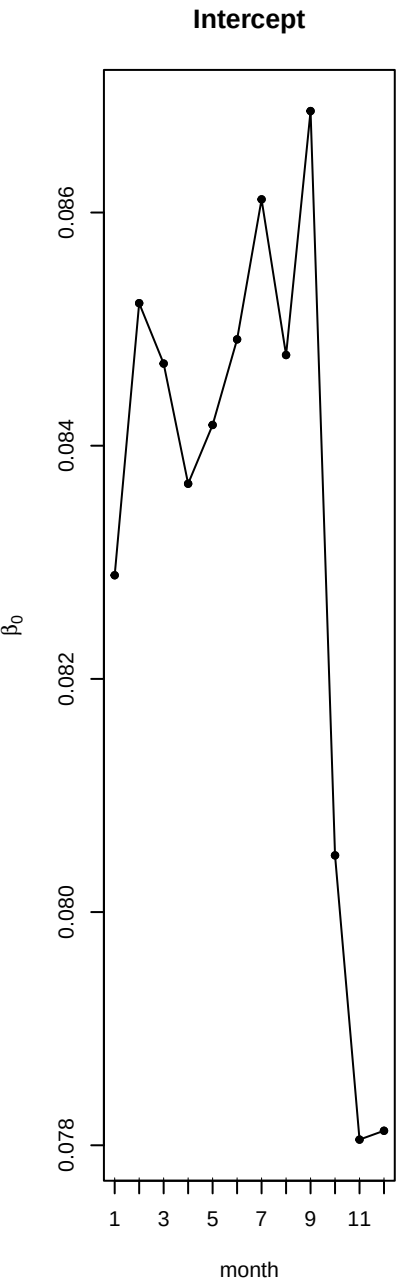

effect of z1
1.0
0.5
0.0
−0.5
−1.0
1 * cos(pi * dat$t)
0.0
0.2
0.4
0.6
0.8
1.0
dat$t
help("bbsc")

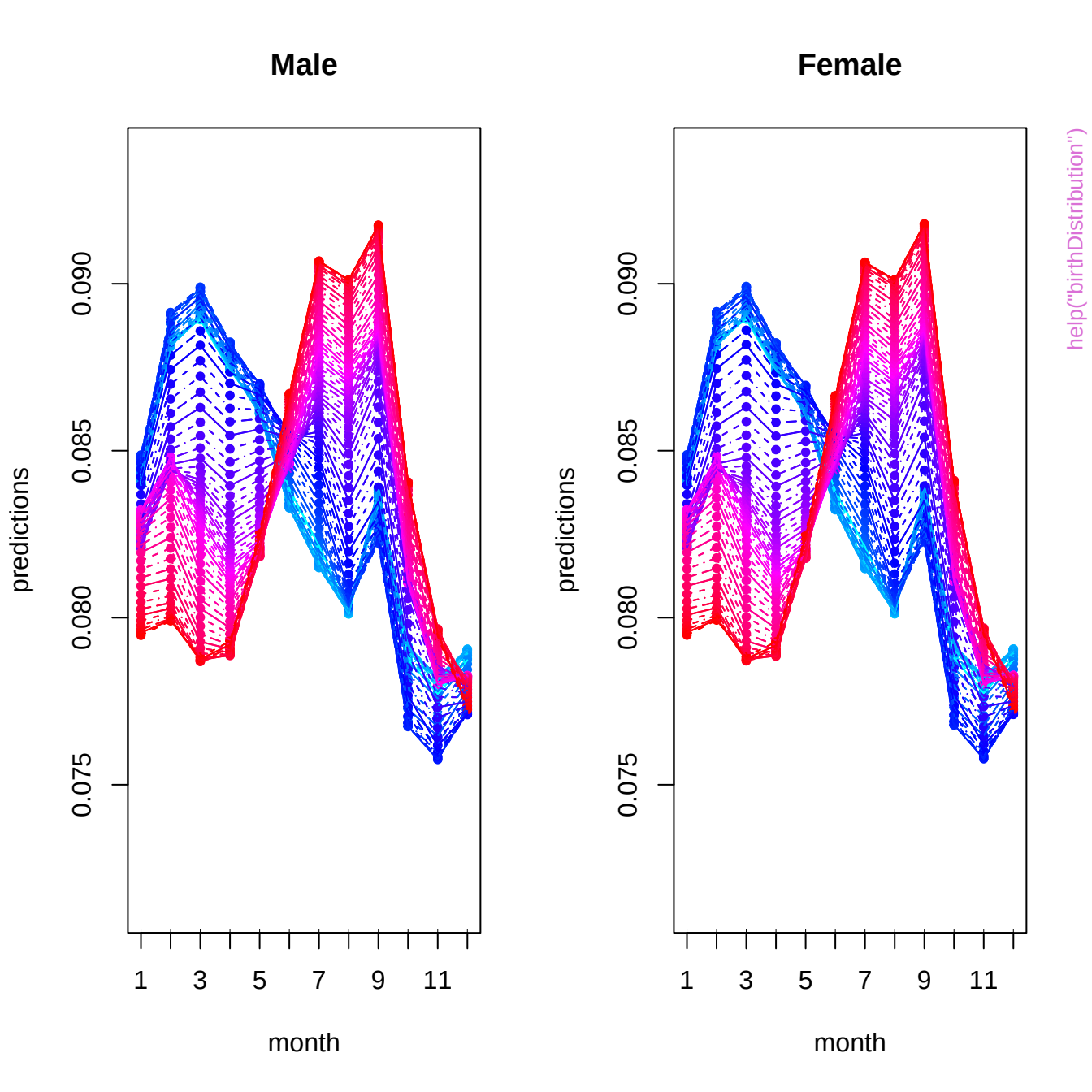

Male
Female
densities
0.075
0.080
0.085
0.090
1
3
5
7
9
11
month
help("birthDistribution")

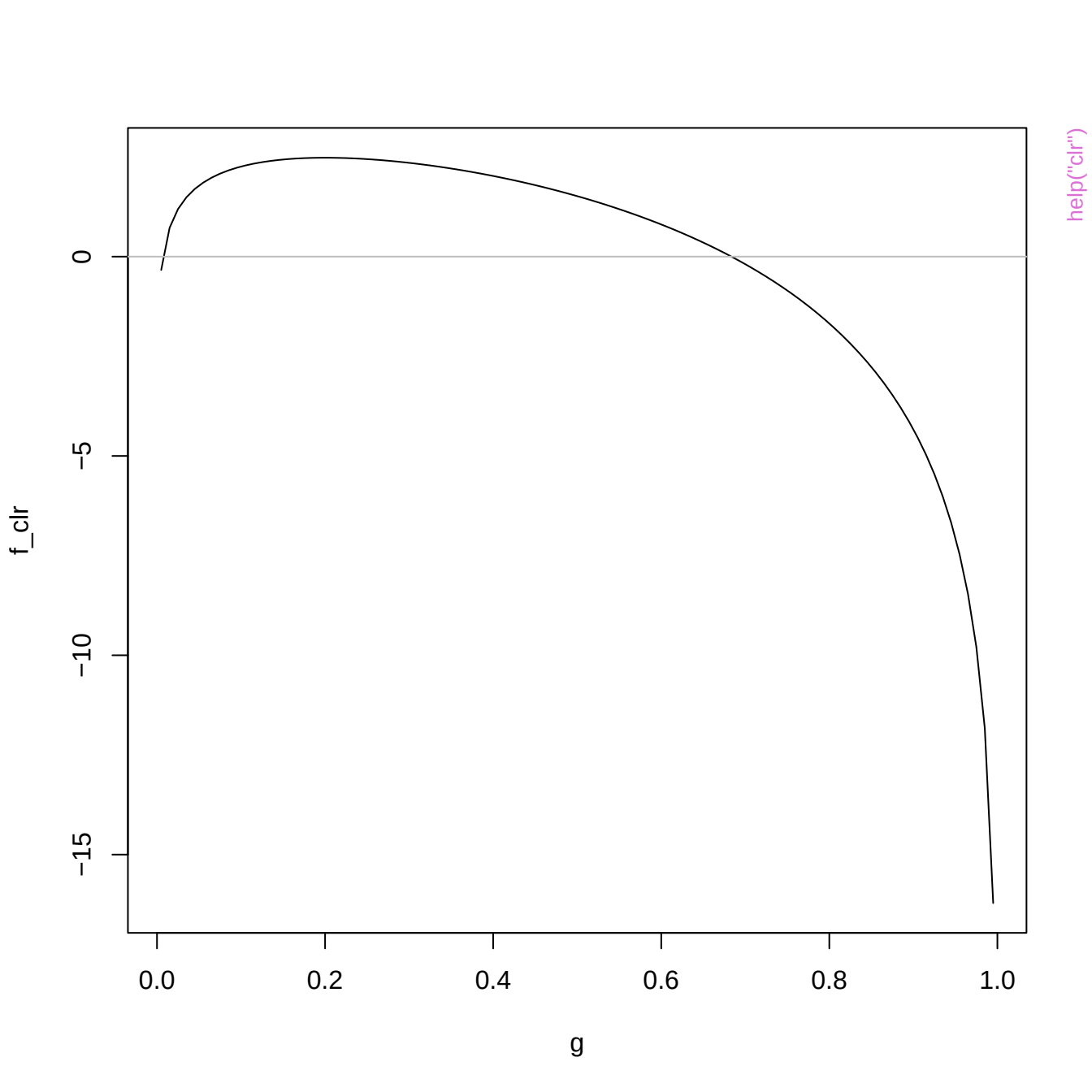

offset + bols(ONEx) %A0% bbsc(mo
coef
0.04
0.02
0.00
−0.02
−0.04
−0.06
2
4
6
8
10
12
month

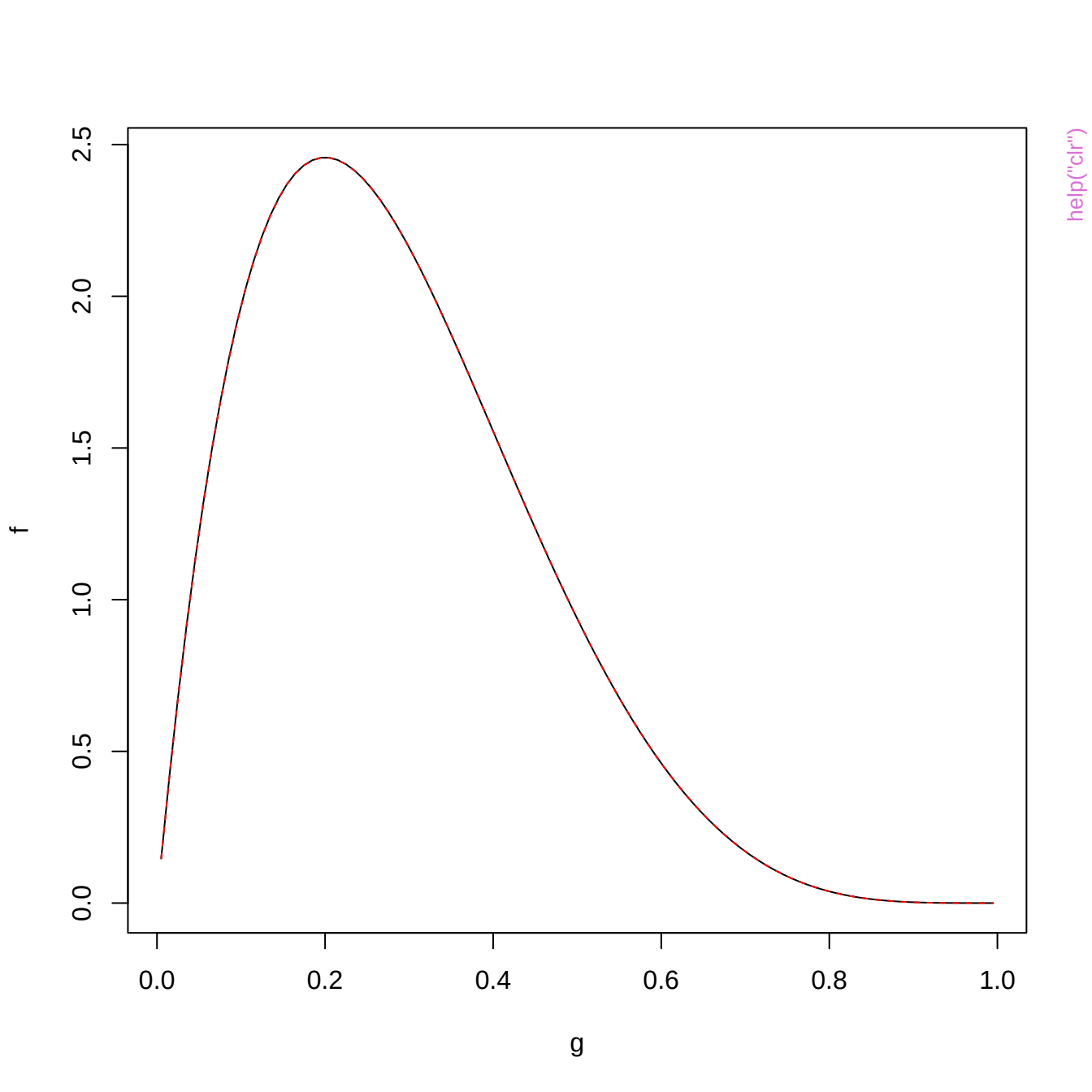

bolsc(sex) %O% bbsc(month)
coef
3e−04
2e−04
1e−04
0e+00
−1e−04
−2e−04
−3e−04
2
4
6
8
10
12
month


bbsc(year) %O% bbsc(month)
year
2020
2010
2000
1990
1980
1970
1960
1950
2
4
6
8
10
12
month
help("birthDistribution")


Intercept
0.086
0.084
0.082
0.080
0.078
β_0
1
3
5
7
9
11
month

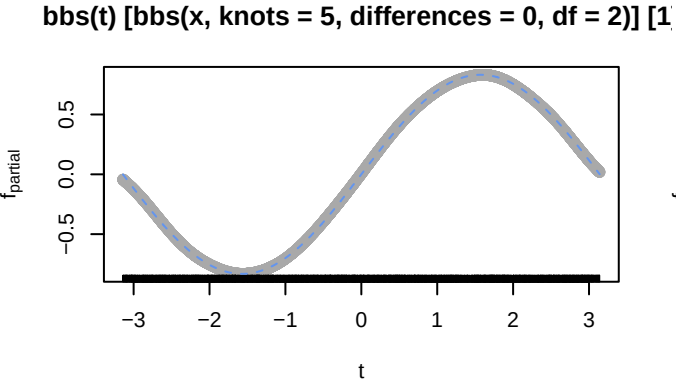

Male
Female
predictions
0.075
0.080
0.085
0.090
1
3
5
7
9
11
month
help("birthDistribution")

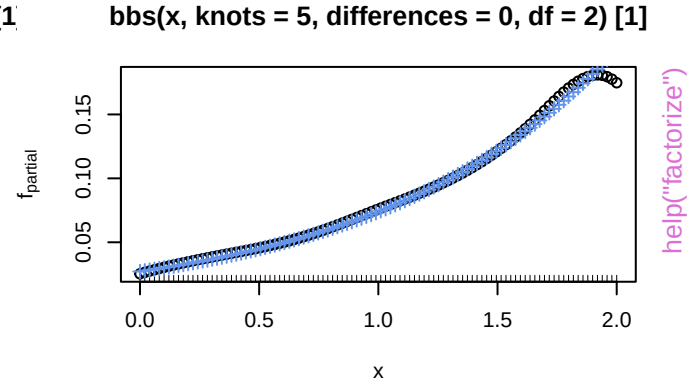

help("clr")
f_clr
0
−5
−10
−15
0.0
0.2
0.4
0.6
0.8
1.0
g

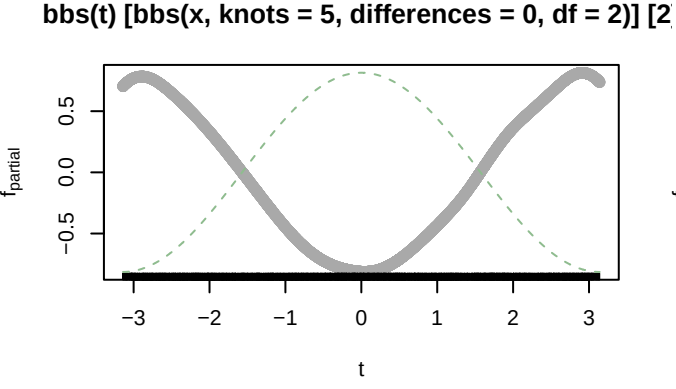


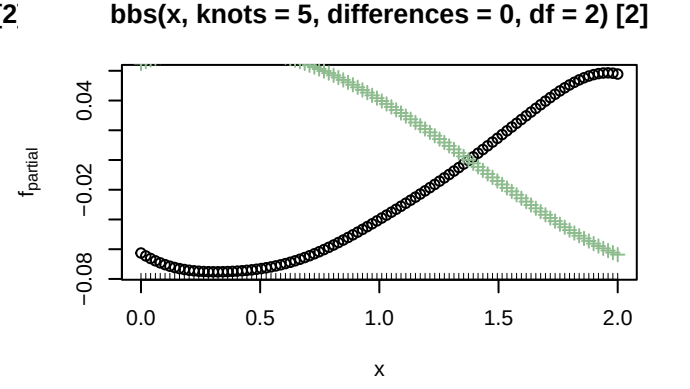
help("clr")

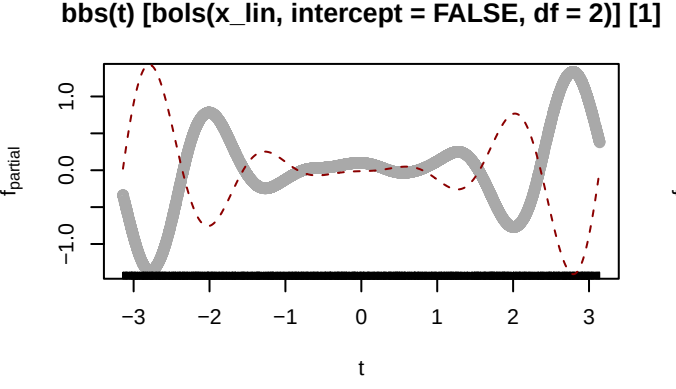
help("factorize")

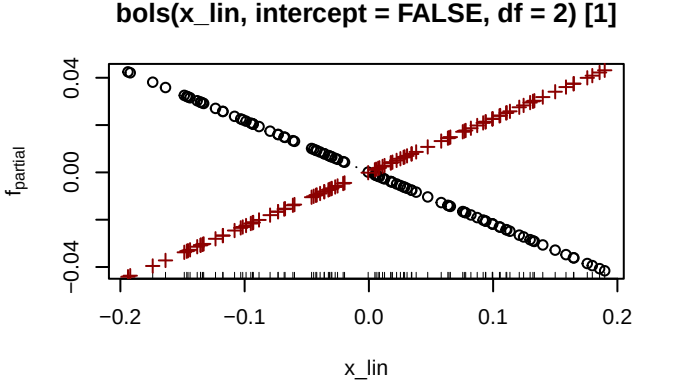

bbs(t) [bbs(x, knots = 5, differences = 0, df = 2)] [1
f_partial
0.5
0.0
−0.5
−3
−2
−1
0
1
2
3
t

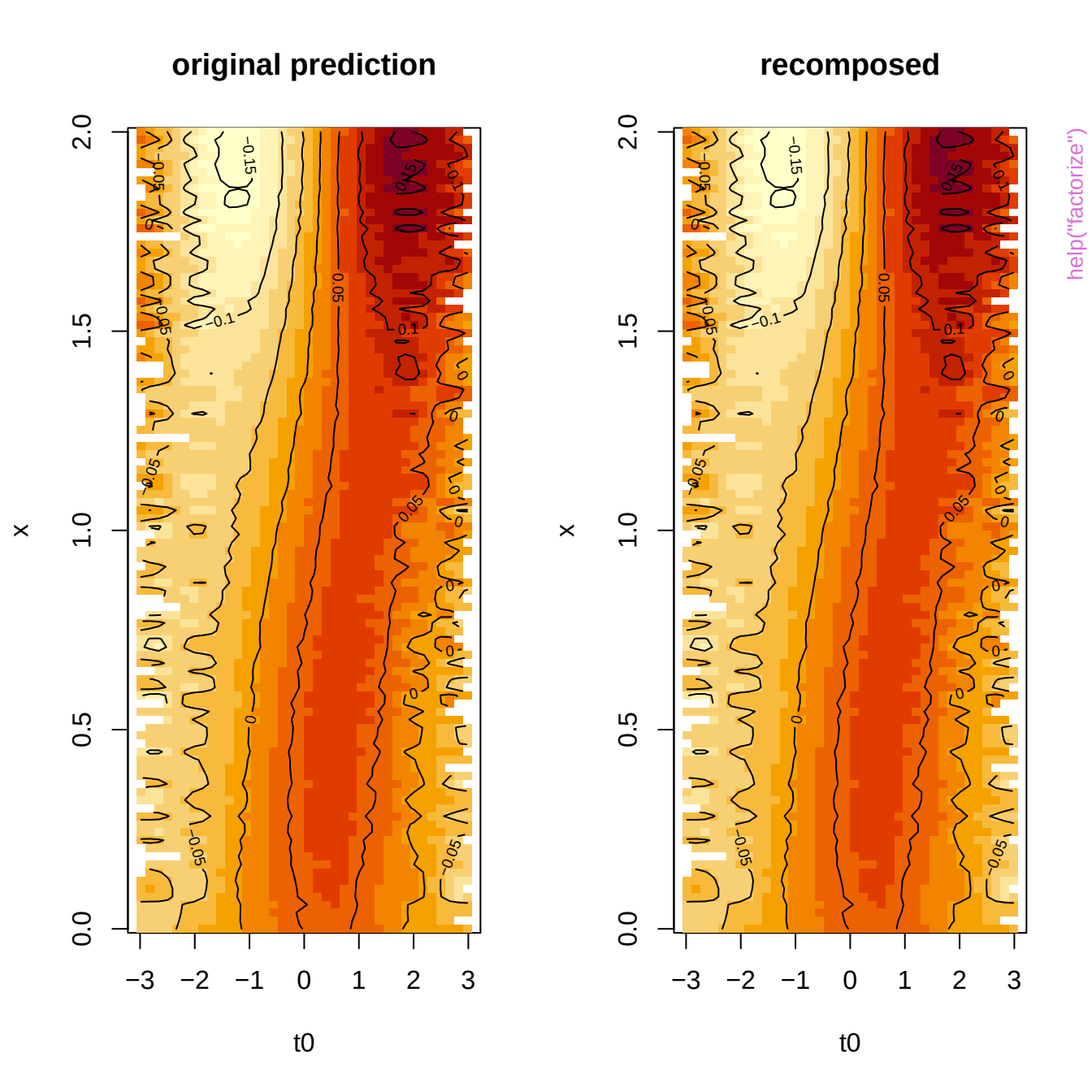

bbs(x, knots = 5, differences = 0, df = 2) [1]
f_partial
0.15
0.10
0.05
0.0
0.5
1.0
1.5
2.0
x
help("factorize")

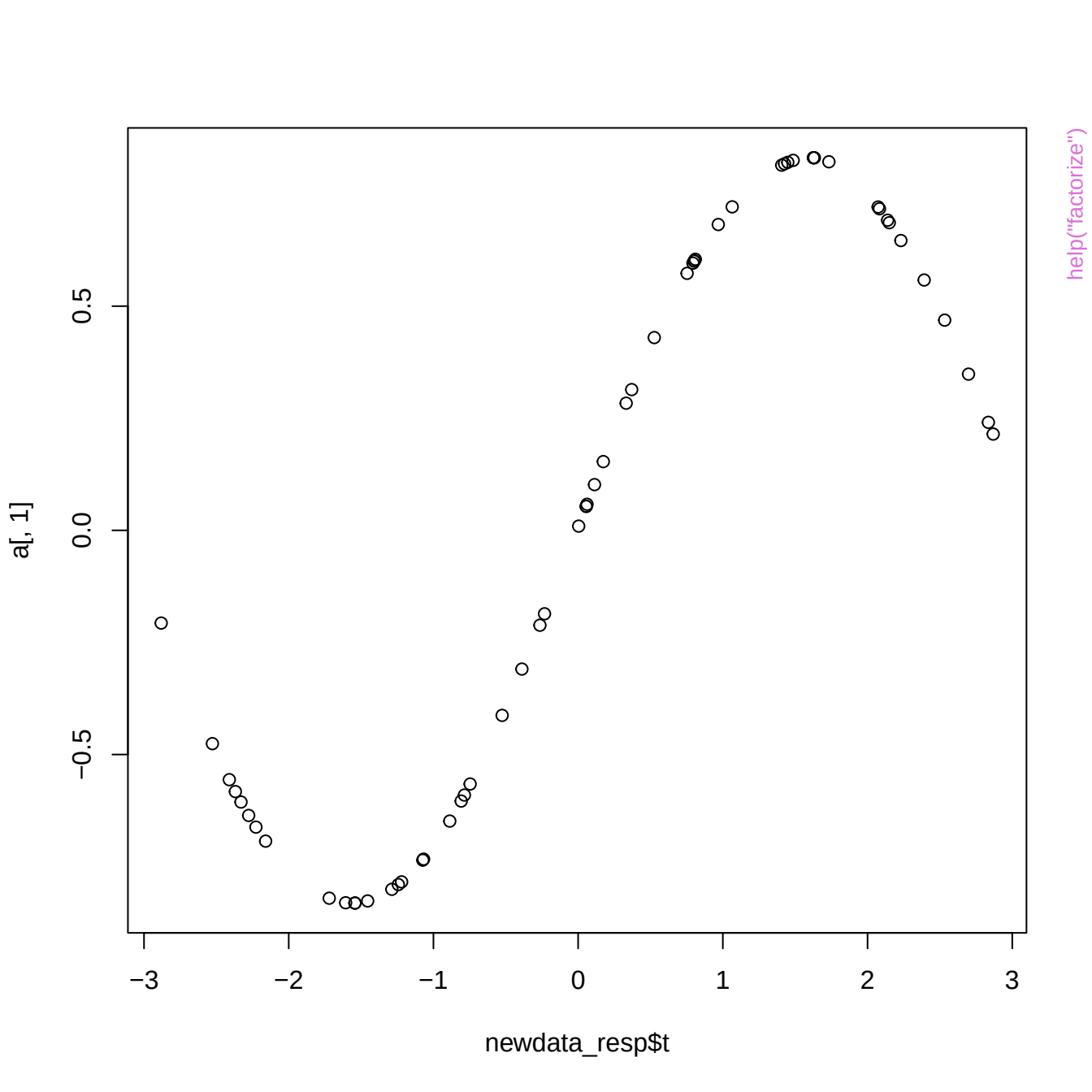

bbs(t) [bbs(x, knots = 5, differences = 0, df = 2)] [2
f_partial
0.5
0.0
−0.5
−3
−2
−1
0
1
2
3
t

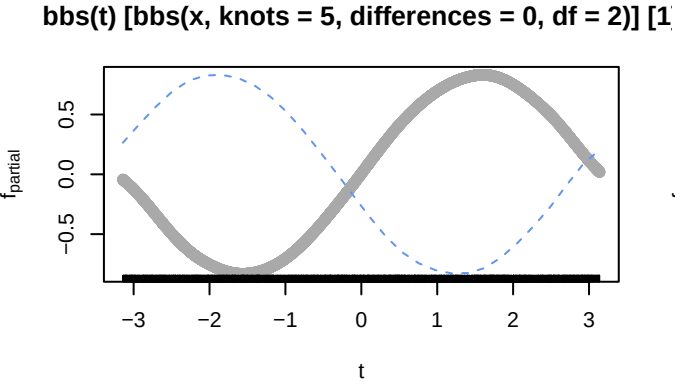

bbs(x, knots = 5, differences = 0, df = 2) [2]
f_partial
0.04
−0.02
−0.08
0.0
0.5
1.0
1.5
2.0
x

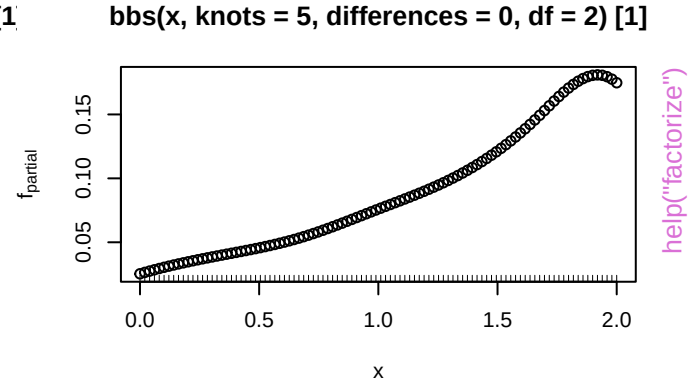

bbs(t) [bols(x_lin, intercept = FALSE, df = 2)] [1]
f_partial
1.0
0.0
−1.0
−3
−2
−1
0
1
2
3
t

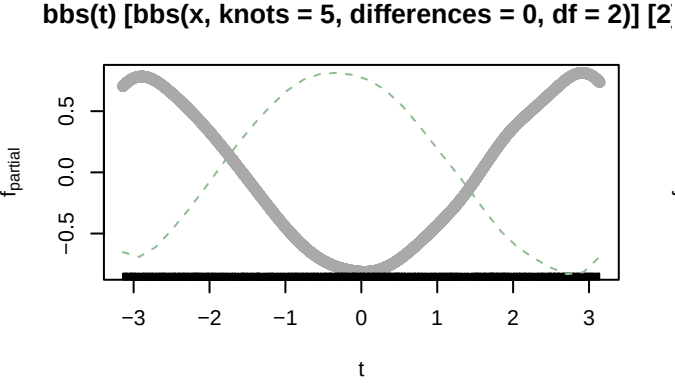

bols(x_lin, intercept = FALSE, df = 2) [1]
f_partial
0.04
0.00
−0.04
−0.2
−0.1
0.0
0.1
0.2
x_lin

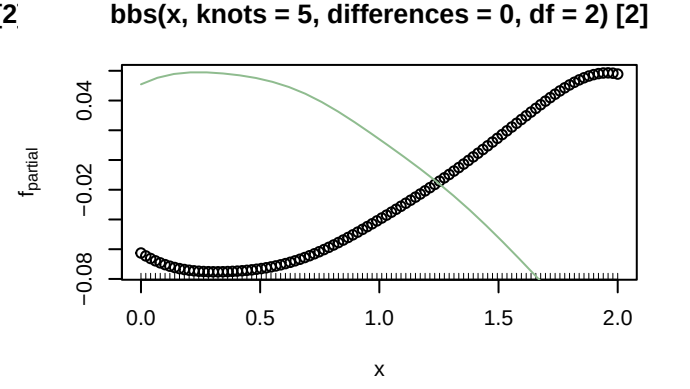

original prediction
recomposed
x
t0
0.0
0.5
1.0
1.5
2.0
−3
−2
−1
0
1
2
3
help("factorize")

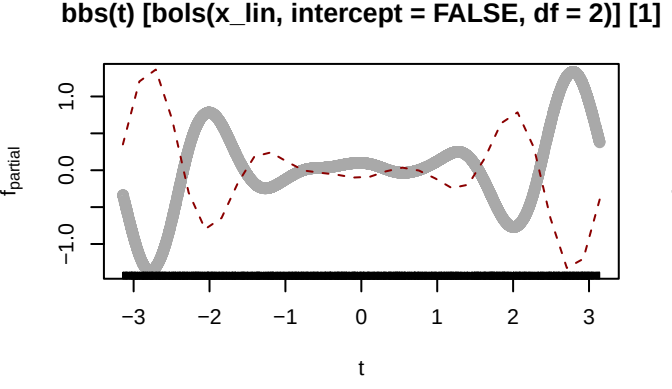

a[, 1]
0.5
0.0
−0.5
−3
−2
−1
0
1
2
3
newdata_resp$t
help("factorize")

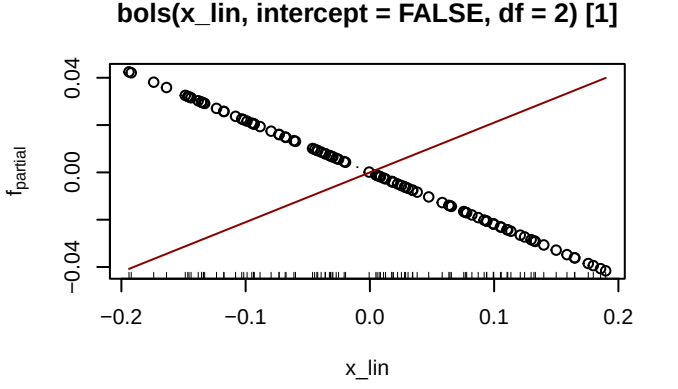

bbs(t) [bbs(x, knots = 5, differences = 0, df = 2)] [1
f_partial
0.5
0.0
−0.5
−3
−2
−1
0
1
2
3
t

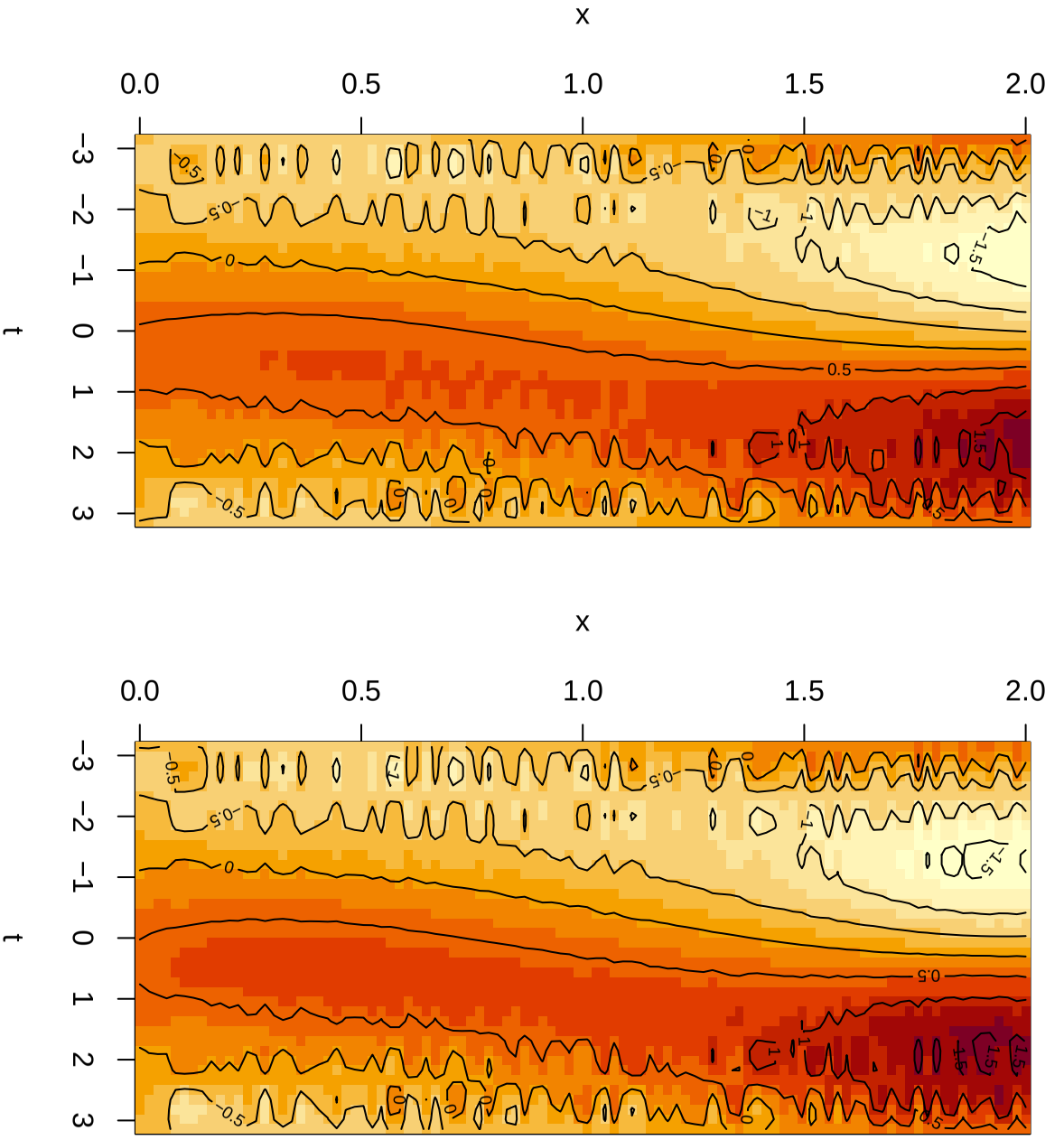

bbs(x, knots = 5, differences = 0, df = 2) [1]
f_partial
0.15
0.10
0.05
0.0
0.5
1.0
1.5
2.0
x
help("factorize")


bbs(t) [bbs(x, knots = 5, differences = 0, df = 2)] [2
f_partial
0.5
0.0
−0.5
−3
−2
−1
0
1
2
3
t

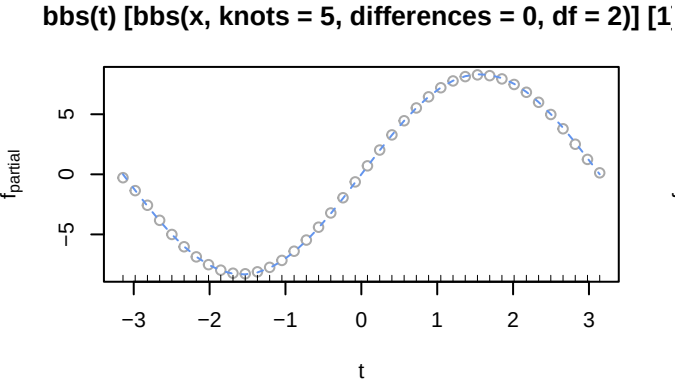

bbs(x, knots = 5, differences = 0, df = 2) [2]
f_partial
0.04
−0.02
−0.08
0.0
0.5
1.0
1.5
2.0
x

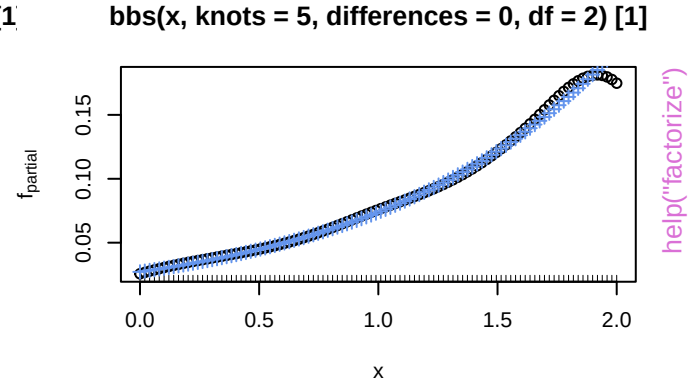

bbs(t) [bols(x_lin, intercept = FALSE, df = 2)] [1]
f_partial
1.0
0.0
−1.0
−3
−2
−1
0
1
2
3
t

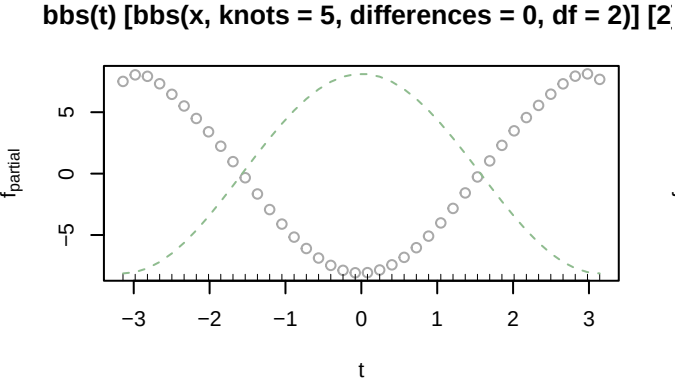

bols(x_lin, intercept = FALSE, df = 2) [1]
f_partial
0.04
0.00
−0.04
−0.2
−0.1
0.0
0.1
0.2
x_lin

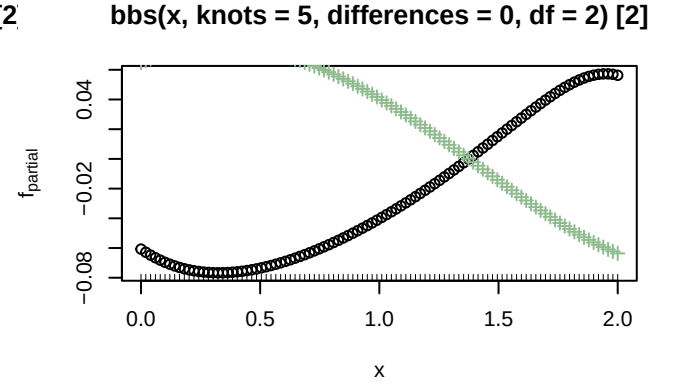


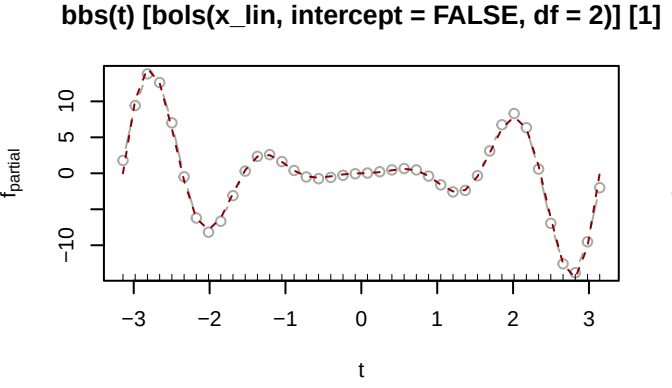
help("factorize")

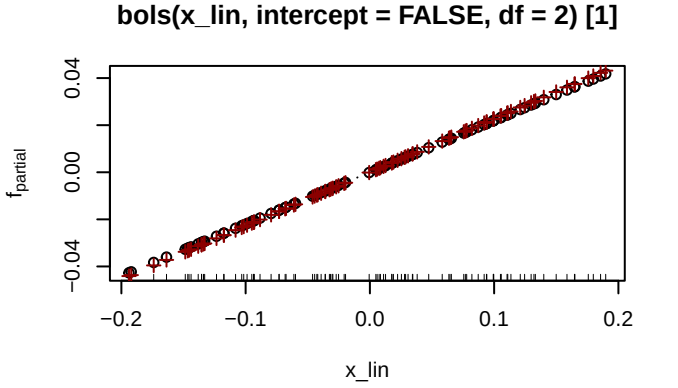




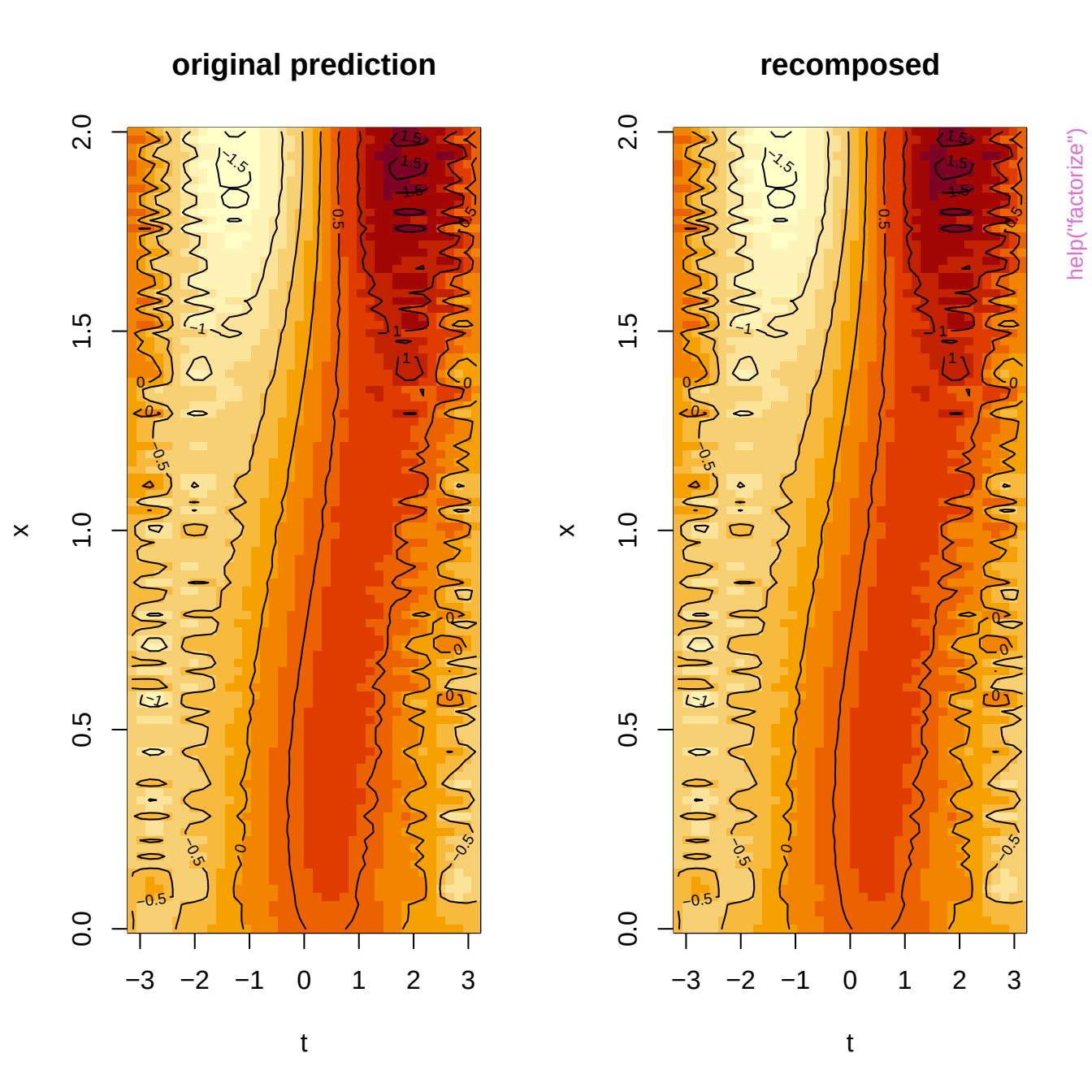








help("factorize")


original prediction
recomposed
x
t
0.0
0.5
1.0
1.5
2.0
−3
−2
−1
0
1
2
3
help("factorize")

help("factorize")